

Lecture 21: Operator Execution

Sorting and Aggregation

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Operator Execution in DBMS

Operator Execution

- DBMS executes queries by composing operators.
- Operators are algorithms that implement relational algebra operations.
- We will focus on sorting and aggregation.

Disk-Oriented DBMS

- Data is stored on disk.
- Data is brought into memory in pages.
- Algorithms must be designed to minimize disk I/O.

Sorting

Need for Sorting

- The relational model is unsorted.
- Sorting is needed for:
 - ORDER BY
 - DISTINCT
 - GROUP BY

In-Memory Sorting Algorithms

- Quicksort
- Mergesort
- Heapsort

Top-N Queries

- Queries with `ORDER BY` and `LIMIT`.
- Example: Find the 10 most expensive products.
- Heapsort is ideal if N elements fit in memory.

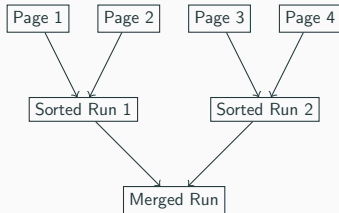
External Merge Sort

- Phase 1: Sorting
 - Read pages into memory.
 - Sort the pages.
 - Write the sorted pages to disk.
- Phase 2: Merging
 - Read sorted pages into memory.
 - Merge the sorted pages.
 - Write the merged pages to disk.

2-Way External Merge Sort

- N pages, B buffer pool pages.
- Number of passes: $1 + \lceil \log_{B-1}(\lceil N/B \rceil) \rceil$
- I/O cost: $2N \times (\text{number of passes})$

2-Way External Merge Sort: Illustration



Problem with More Pages

- If we have more pages than buffer pool pages, we need to do more passes.
- This increases the I/O cost.

Double Buffering

- Use two buffers for each run.
- While one buffer is being used for merging, the other is being filled from disk.
- This can reduce the I/O cost.

Thank You
Questions?